

CipherPlay

AI-Powered Privacy-First Scavenger Hunt for Children

1. Project Summary

CipherPlay is an intelligent mobile application designed to encourage children to stay physically active while enjoying a fun gaming experience. Instead of keeping users seated in front of a screen, the application motivates them to move around their surroundings by completing object-finding challenges.

The system uses artificial intelligence to recognize everyday household items and create interactive missions. Children earn points by locating and capturing the required objects within a specified time. To ensure privacy, all captured images are protected through advanced on-device AES-256 encryption using a novel physical entropy source — bubble coordinates extracted from an aquarium via OpenCV — before being stored or shared.

By combining AI, mobile gaming, and cybersecurity, CipherPlay provides a safe and engaging environment that promotes healthy habits and responsible technology use.

2. Abstract

Mobile technology has become deeply embedded in children's daily lives, leading to a significant rise in sedentary behaviour and reduced physical activity among young age groups. Existing mobile games provide entertainment but fail to bridge the gap between screen time and physical engagement. Simultaneously, applications that access a device's camera raise serious privacy concerns, particularly when sensitive home environments are captured and transmitted online. Current encryption solutions rely on computationally generated pseudo-random numbers which, while functional, are theoretically vulnerable due to their predictable nature.

This project proposes CipherPlay, an AI-powered mobile scavenger hunt game designed specifically for children aged 4 to 14. The application uses YOLO v8-based real-time object detection to scan a child's room and identify available objects, then challenges the child to physically locate and photograph those objects within a time limit.

To address privacy concerns, CipherPlay introduces a novel low-cost physical entropy-based encryption system. Instead of relying on expensive hardware random number generators (HRNG), the system uses continuously moving bubble coordinates from an aquarium tank tracked by a secondary camera feed. OpenCV extracts real-time bubble coordinates which serve as a true entropy source for generating AES-256 encryption keys. Every photo captured during gameplay is encrypted on-device before any data leaves the application.

This project makes three distinct research contributions: a gamified physical activity system for children using computer vision, a novel low-cost physical entropy source as an alternative to commercial HRNG devices, and a real-time on-device encryption pipeline for child data privacy.

3. Problem Statement & Gap Analysis

3.1 Health Risks of Sedentary Screen Time

- Children aged 4-14 average 4-6 hours of daily passive screen time, leading to childhood obesity and impaired motor development.
- Prolonged stationary activities like scrolling and static gaming cause a significant reduction in essential social interaction.
- Current mobile app ecosystems actively reward sedentary behaviour, offering zero incentives for physical movement or exercise.

3.2 Domestic Privacy Risks in Camera-Enabled Apps

- Camera-enabled applications capture highly sensitive data, including home layouts and private living spaces, during indoor use.
- Low awareness of data rights in developing nations leads to installation of apps with weak encryption and high interception risks.
- The children's app market currently lacks a robust on-device encryption standard to protect family data from misuse.

3.3 Economic Barriers to Secure Encryption

- True Random Number Generation (TRNG) is vital for security, yet hardware generators are financially inaccessible to researchers in developing countries.
- Developers are forced to use Pseudo-Random Number Generators (PRNGs), which are mathematically deterministic and predictable.
- The high cost of physical hardware undermines the entire security infrastructure of small-scale applications.

3.4 Gap Analysis

No existing mobile application for children combines physical activity gamification, AI-powered object detection, and on-device encryption with a physical entropy source. The technologies to solve all three problems exist independently but have never been combined into a single cohesive system. This project fills that gap.

4. Proposed Solution

CipherPlay is a mobile application for Android that transforms a child's physical environment into a dynamic game arena. The application has two distinct but integrated systems operating simultaneously: the gameplay layer and the security layer.

4.1 Gameplay Layer

- AI-Driven Detection: Uses an on-device YOLO v8 model to identify household objects in real time, turning the physical environment into a digital playground.

- **Physical Activity:** Promotes movement by challenging children to locate and photograph specific objects within a countdown timer, shifting screen time from passive to active.
- **Smart Verification:** Employs a CNN-based model to verify captured photos and a lightweight ML model to automatically adjust difficulty based on the child's age and performance.

4.2 Security Layer

- **Nature-Based Entropy:** Captures the chaotic, unpredictable movement of bubbles in an aquarium via OpenCV to generate True Random Numbers for encryption.
- **On-Device AES-256:** Every gameplay photograph is encrypted immediately on-device using a seed derived from bubble coordinates, ensuring raw images never reach the server.
- **Intercept-Proof Privacy:** By using a unique physical entropy key, the system ensures that even intercepted data remains undecipherable to unauthorized actors.

5. Main Features

Feature	Description
AI-Powered Object Recognition	YOLO v8 on-device real-time detection of household objects
Interactive Scavenger Hunt	Physical activity-based challenges that incentivize movement
Real-Time Challenge Generation	Dynamic mission creation based on detected objects in room
CNN Photo Verification	Automatic confirmation of correct object identification
Adaptive Difficulty (ML)	Auto-adjusts based on child age and performance history
Physical Entropy Encryption	Aquarium bubble coordinate tracking as true randomness source
AES-256 On-Device Encryption	All images encrypted before any data transmission
Parent Monitoring Dashboard	Activity reports, time played, privacy settings
Rewards & Achievement System	Badges, streaks, speed stars, PIXEL costumes, story unlocks

6. Objectives

- Use on-device YOLO v8 AI to turn sedentary screen time into a movement-based game that challenges children to find and scan real-world objects.
- Implement a lightweight ML model that automatically adjusts game difficulty and physical tasks based on the child's age and historical performance.
- Replace expensive hardware with a bubble-tracking entropy source to generate true random keys for local AES-256 encryption.
- Ensure child privacy by encrypting all images on-device and validating key randomness through NIST SP 800-22 industry benchmarks.
- Deliver a fully functional Android application with a transparent parent dashboard and engaging PIXEL character-driven story.

7. Target Audience

Primary Users

Children aged 4-14 years playing under parental supervision at home.

Secondary Users

Parents seeking healthy, safe, and physically active digital alternatives to passive screen time for their children.

Institutional Users

Schools and educational organizations interested in activity-based learning solutions and gamified physical activity tools. Particular relevance in Pakistan where childhood obesity rates are rising and digital literacy around data privacy remains low.

8. Tools & Technologies

Domain	Technology / Tool
Game Development	Unity / Android Studio (Java, Kotlin)
Object Detection (AI)	YOLO v8 (TensorFlow Lite) — on-device mobile deployment
Object Verification	Convolutional Neural Network (CNN)
Computer Vision	OpenCV — Bubble coordinate tracking
Encryption	AES-256 (Physical entropy key seeding)
Entropy Validation	NIST SP 800-22 Statistical Test Suite
Backend / Database	Firebase
Hardware (Entropy)	Aquarium-based Bubble Tracking System (~Rs. 2,000)
Project Management	GitHub, Trello

9. Game Design & Story

9.1 Game Overview

CipherPlay is a single-player mobile scavenger hunt game designed for children aged 4 to 14. The game takes place entirely inside the child's real physical environment. There are no fictional worlds or virtual levels. The child's own room IS the game world, making every playthrough unique and personal.

9.2 Story & Theme

A friendly robot named PIXEL has landed in the child's house and lost all of his memory chips. Each memory chip is disguised as a common household object. PIXEL needs the child's help to find and recover all his lost chips before time runs out. Every object the child finds and photographs gives PIXEL one memory back. As PIXEL recovers more memories, he unlocks new abilities, tells funny stories, and reveals hidden surprises.

9.3 Characters

PIXEL (Main Character): A small round friendly robot with big expressive eyes and small antenna. PIXEL reacts emotionally to everything the child does — celebrates when an object is found, looks worried when timer runs low. He never criticizes the child for failing.

The Parent (Background Character): Not a visible in-game character but has a dedicated dashboard for monitoring activity reports, time played, steps taken, and all privacy settings.

The Bubble Guardian (Hidden Character): A secret character children discover after completing enough rounds. Described as the keeper of secrets who scrambles photos so only PIXEL and the child can see them — a child-friendly explanation of the encryption system.

9.4 Game Levels & Progression

Level	Name	Age Group	Description
Level 1	Memory Sprout	4-6 years	Very common object, 45 sec timer, large text and icons
Level 2	Memory Scout	7-9 years	Common objects, 30 sec timer, moderate difficulty
Level 3	Memory Hunter	10-12 years	Less obvious objects, 20 sec timer, specific description
Level 4	Memory Master	13-14 years	Specific attributes required, 15 sec timer, chain challenges

Table 1: Game Levels

9.5 Reward & Motivation System

- Memory Chips — collected after each successful find, shown filling up PIXEL's body
- Daily Streak Badges — awarded for playing multiple days in a row
- Speed Stars — bonus stars for finding objects very quickly
- PIXEL Costumes — unlockable outfits earned through milestones
- Story Unlocks — PIXEL reveals funny memories as the child progresses

10. Expected Outcomes & KPIs

10.1 Product Outcome

- A fully functional Android application combining real-time AI and adaptive gameplay.
- Core features include on-device object detection, physical entropy-based encryption, and a dedicated parent dashboard.

10.2 Research Outcome

- A validated aquarium-bubble tracking system as an affordable alternative to commercial HRNGs (hardware cost under Rs. 2,000).
- NIST SP 800-22 test suite results confirming hardware-grade cryptographic randomness quality.

10.3 Key Performance Indicators (KPIs)

KPI	Target
YOLO v8 Object Detection Accuracy	Above 85% on household objects
NIST SP 800-22 Statistical Tests	All 15 tests passed for bubble-generated keys
Physical Movement per Session	Minimum 50 steps (measured by accelerometer)
On-Device Encryption Speed	Under 200ms per image
Hardware Entropy Cost	Under Rs. 2,000 total
Parent Satisfaction Rate	Over 80% positive in user testing

11. Project Timeline (2 Semesters)

Phase	Timeline	Tasks
Phase I Setup & Design	Weeks 1-6 (Sep-Oct 2026)	Requirements & setup, Literature review, UI/UX wireframes, Hardware entropy system setup & calibration
Phase II Core Development	Weeks 7-21 (Oct 2026 - Feb 2027)	YOLO v8 integration, CNN verification model, Physical entropy OpenCV pipeline, AES-256 encryption, NIST validation, Game loop development
Phase III Integration & Testing	Weeks 22-27 (Feb-Apr 2027)	Adaptive difficulty ML model, Parent dashboard, Full system integration, User testing with 20+ users, Bug fixing
Phase IV Documentation	Weeks 28-33 (Apr-Jul 2027)	Technical report, Research contribution write-up, Final presentation and submission

Table 2: Project Phases

12. Completion Milestones

Milestone	Timeline	Completion Criteria
M1 — Requirements & Setup	Weeks 1-2 (Sep 2026)	Tech stack confirmed, hardware purchased, GitHub repo created
M2 — Literature & UI/UX	Weeks 3-5 (Oct 2026)	Literature survey complete, UI wireframes finalized
M3 — AI Object Detection	Weeks 6-9 (Oct-Nov 2026)	YOLO v8 on-device, tested on 10+ objects, >85% accuracy
M4 — Physical Entropy	Weeks 10-13 (Nov-Dec 2026)	Bubble tracking operational, NIST randomness tests run
M5 — Encryption Pipeline	Weeks 14-17 (Dec 2026 - Jan 2027)	AES-256 integrated, encryption time <200ms per image
M6 — Game Development	Weeks 18-21 (Jan-Feb 2027)	Full game loop: scan, challenge, timer, verify, points
M7 — Parent Dashboard	Weeks 22-24 (Feb-Mar 2027)	Dashboard with activity reports and privacy controls
M8 — User Testing	Weeks 25-27 (Mar-Apr)	Tested with 20+ real users, top

	2027)	improvements implemented
M9 — Documentation	Weeks 28-30 (Apr-May 2027)	Full technical report, research write-up, README committed
M10 — Final Submission	Weeks 31-33 (May-Jul 2027)	Final presentation, code, docs, video demo submitted

Table 3: Completion Criteria

13. Challenges & Mitigations

13.1 Time Management

The project spans three technical domains: game development, artificial intelligence, and cryptography. Tasks will be divided by team member specialization and tracked weekly using GitHub commits and bi-weekly supervisor meetings.

13.2 Technical Skills to Learn

- YOLO v8 on-device deployment via Ultralytics official documentation and TensorFlow Lite mobile guides.
- Physical entropy extraction using OpenCV developed through fluid dynamics research and bubble detection tutorials.
- AES-256 encryption implementation following NIST cryptographic standards documentation.
- NIST SP 800-22 test suite applied to validate randomness quality of bubble-generated keys.

13.3 Hardware Setup

The bubble entropy system requires physical hardware including an aquarium, air pump, and secondary camera. This will be set up and calibrated in the first two weeks and tested for consistent randomness output before integration with the encryption pipeline. Total estimated cost: under Rs. 2,000.

14. Literature Review

The problem of childhood physical inactivity due to mobile screen time is well documented in pediatric health literature. Studies consistently show that children who spend more than two hours per day on screens exhibit lower physical fitness, higher body mass index, and reduced attention span compared to peers with limited screen time [7].

In the domain of computer vision, YOLO v8 released by Ultralytics in 2023 [1] achieves high accuracy with low computational requirements, making it suitable for on-device mobile deployment through TensorFlow Lite [3]. Prior work on indoor object detection has focused primarily on assistive technology for visually impaired users, and no existing application has applied YOLO-based room scanning to children's gameplay.

In cryptography, true random number generation is recognized as foundational to strong encryption. Hardware Random Number Generators (HRNG) using physical phenomena are considered cryptographically superior to software-based Pseudo-Random Number Generators (PRNG) [2]. Cloudflare's widely publicized use of a wall of lava lamps as an entropy source demonstrates the validity of physical fluid dynamics as a randomness source [5]. However, no existing research has explored aquarium bubble dynamics as a low-cost alternative for mobile application security.

Technology / Approach	Limitation	In CipherPlay
Pokemon Go / AR Games	Requires GPS and outdoor space; not suitable for young children indoors	No
YOLO Object Detection	Not previously applied to children's game contexts	Yes
Software PRNG Encryption	Not truly random, theoretically predictable	No
Commercial HRNG Devices	Cost tens of thousands of rupees, inaccessible for small developers	No
Cloudflare Lava Lamp Wall	Industrial scale, not adapted for mobile consumer use	No
Bubble Entropy (Ours)	Novel low-cost physical entropy source for mobile AES-256 encryption	Yes
CNN Verification Models	Not previously used for anti-cheat in children's activity games	Yes
Adaptive Difficulty ML	Not applied to scavenger hunt physical activity contexts	Yes

Table 4: Comparative Analysis

15. Project Success Criteria

The final result will be a working Android application demonstrating a complete scavenger hunt game loop driven by real-time AI object detection, fully encrypted on-device image handling using physical entropy keys, and a transparent parent dashboard.

Success will be proved by:

- YOLO v8 object detection accuracy above 85% on household objects.
- All 15 NIST SP 800-22 statistical tests passing for bubble-generated key sequences.

- On-device AES-256 encryption completing in under 200 milliseconds per image.
- Positive user feedback from at least 80% of parent testers.
- The research contribution of a sub-Rs. 2,000 physical entropy source documented and made available as an open-source library.

16. References

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